Supplementary materials

1 RQ2

Table 1: Regularity of observed constraints with $N_{ij} \geq \tau$: mean number k of significant relations ($w_{ij}(r) \geq \tau_c = 0.1$), share of constraints with at least two significant relations, and mean mass m left outside the dominant relation.

Dataset	τ	$\lvert\mathcal{O}_\tau\rvert$	Mean k	$k \geq 2$ (%)	Mean m (%)
Country	1	26 344	1.49	39.6	15
	10	14 282	1.64	53.5	20
	59	7 436	1.66	58.2	22
	110	5 748	1.67	60.2	23
	160	4 898	1.68	60.7	23
	190	4 506	1.69	61.6	23
Musical group	1	3 902	1.42	30.2	13
	10	970	1.73	47.2	21
	30	478	1.80	47.7	22
	60	294	1.76	46.3	22
	100	240	1.78	46.7	23
	150	190	1.69	41.1	21
Politician	1	19 236	1.52	36.4	15
	14	5 788	1.85	58.1	26
	42	3 394	1.81	56.8	26
	128	1 848	1.82	57.8	26
	389	988	1.82	57.1	26
	1185	480	1.77	55.0	25
WD27M	1	1 586	1.88	58.5	25
	10	1 034	2.00	67.9	29
	100	542	1.95	67.5	31
	1000	236	1.81	59.3	27
	10 000	72	1.44	33.3	17

2 More Fact Checking Results

2.1 Fact checking using Soulard et al. split

Table 2 shows fact-checking results using the split protocol described by Soulard et al.

2.2 Difference between *TCL*and *TCL-NP*

3 Domain size distribution (RQ2)

We address two questions: (1) Is there a consensus regarding the temporality of the observed facts? (2) Are the inferred domains informative?

Metric	TCL	PaTeCon	Soulard et al.
Musical group			
Coverage	**75.4**	48.7	38.2
Correct / total	**53.0**	31.8	23.8
Invalid facts detected / total	**19.9**	8.4	3.9
P_+	**65.1**	59.1	58.5
R_+	88.0	97.0	**98.3**
$F1_+$	**74.8**	73.5	73.4
P_-	81.5	**92.1**	**92.1**
R_-	**52.7**	34.2	21.8
$F1_-$	**64.0**	49.9	35.3
Mean F1	**69.4**	61.7	54.3
Country			
Coverage	**100.0**	20.6	13.9
Correct / total	**64.8**	15.9	10.9
Invalid facts detected / total	**41.1**	6.5	3.3
P_+	72.8	69.8	**78.4**
R_+	47.4	**94.0**	89.1
$F1_+$	57.4	80.1	**83.4**
P_-	61.0	**91.4**	77.7
R_-	**82.3**	61.2	60.9
$F1_-$	70.1	**73.3**	68.3
Mean F1	63.7	**76.7**	75.8
Politician			
Coverage	**84.8**	74.9	45.8
Correct / total	**58.0**	47.1	28.0
Invalid facts detected / total	**21.3**	20.7	5.0
P_+	**63.5**	57.3	56.9
R_+	86.4	76.6	**98.7**
$F1_+$	**73.2**	65.5	72.2
P_-	78.7	72.0	**94.1**
R_-	50.3	**51.3**	22.1
$F1_-$	**61.4**	59.9	35.8
Mean F1	**67.3**	62.7	54.0

Table 2: Results of the fact validation task (in %).

	Musical group		Country		Politician		WD27M	
Metric	no prop.	prop.	no prop.	prop.	no prop.	prop.	no prop.	prop.
Facts using inferred constraints (%)	–	69.1	–	34.1	–	33.4	–	94.2
Facts decided by inferred constraints only (%)	–	7.3	–	0.0	–	4.8	–	63.4
Accuracy on facts decided by inferred constraints only (%)	–	72.5	–	–	–	79.4	–	93.5
Inferred constraints right on disagreement (%)	–	61.3	–	78.8	–	55.1	–	91.4
P(invalid | violated), observed	0.90	0.90	0.62	0.62	0.73	0.73	0.92	0.92
P(invalid | violated), inferred	–	0.95	–	0.53	–	0.88	–	0.99
P(valid | satisfied), observed	0.62	0.62	0.59	0.59	0.56	0.56	0.60	0.60
P(valid | satisfied), inferred	–	0.76	–	0.75	–	0.65	–	0.83

Table 3: Use and reliability of inferred constraints on the TCL test_inf split (usage and accuracy in %, probabilities in $[0, 1]$). Inference usage metrics are reported for propagation only. Accuracy on inferred-only decisions is computed over those decisions; disagreement accuracy measures how often inferred constraints are correct when observed and inferred constraints disagree. TCL configurations: min150 for Musical group, min190 for Country, min1185 for Politician, and min100 for WD27M.

(1) The left part of each plot in Figure 1 shows that observed domains are concentrated on small sizes: domains of size 1 to 3 account for 76.2% of pairs for Country and 81.2% for Politician, and domains of size 1 to 2 for 81.3% of pairs for Musical group. Part of these singletons stems from rarely observed pairs, whose domains are mechanically narrow: at the first non-zero threshold, the proportion of singletons drops from 46.6% to 21.4% for Country, from 54.4% to 15.8% for Politician, and from 62.4% to 25.6% for Musical group. Yet even at the highest threshold, where the median domain size rises to 7 for Country, 9 for Politician and 4 for Musical group, domains remain far from the universal relation (i.e. $|\mathcal{L}|$) for Country and Musical group (7.8% and 3.2% of size-13 domains, respectively), whereas Politician shows a larger share (32.5%). The data therefore carry an implicit consensus on the valid temporal relations between pairs of properties, which is not merely a consequence of their rarity, albeit weaker for Politician.

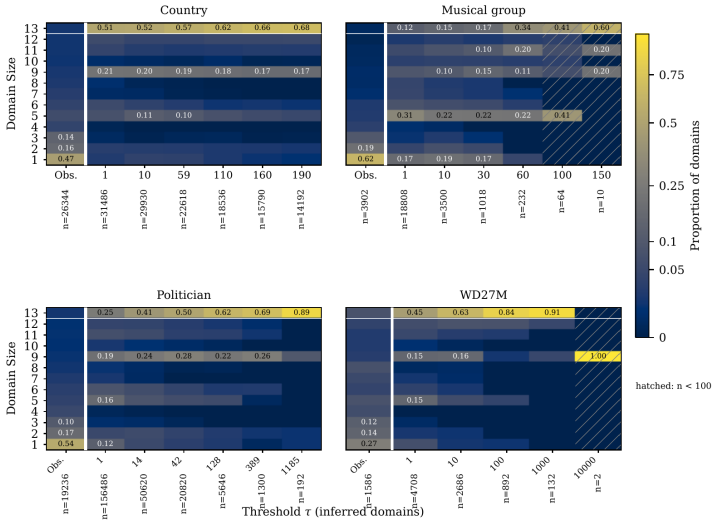


Figure 1: Observed and inferred domain size distributions w.r.t. τ

(2) The right part of each plot in Figure 1 shows the size distribution of inferred domains, i.e., domains of pairs absent from the observations. Domains containing all 13 relations form the largest group, and their proportion grows as the threshold τ increases. This is expected: since our propagation is non-destructive, domains tend to gain relations when several composition paths are available. As τ increases, it filters out rare property pairs, which have narrow domains, and retains the more frequent pairs, which have wide domains. Although compositions $r \circ r'$ leading to the universal relation are discarded, the non-destructive union of partial compositions derived from wide domains is often sufficient to cover all 13 relations.

A support size of 13 does not imply that a domain is uninformative: since propagation is non-destructive, relations within the same domain retain differentiated scores. At most 0.09% of universal domains exhibit a uniform score across all 13 relations for Country and Politician (2.21% for Musical group), and only at the lowest thresholds. Nearly all universal domains present at least 3 score levels at τ_0 (99.5% for Country, 98.0% for Politician, 80.9% for Musical group).

Across all thresholds, the average number of distinct score levels remains stable (between 5.01 and 5.60 for Country, 4.77 and 5.14 for Politician, and 3.55 and 4.66 for Musical group) Detailed per-threshold statistics are provided alongside the code. Thus, inferred domains generally retain differentiated confidence scores, even when they contain all 13 relations.